An Estimate of the Store Size Necessary for Dynamic Storage Allocation

J. M. ROBSON

Oxford University, Oxford, England*

ABSTRACT. Dynamic storage allocation using fixed blocks is usually inefficient in its use of store. The amount of store needed depends on the allocation strategy used. It is proved that for any strategy the amount of store needed is bounded below by a function which rises logarithmically with the size of blocks used. A certain strategy is shown to exceed this bound by a factor of at most 13/4. The exact amount of store needed is found for the case when blocks have sizes 1 and 2 only.

KEY WORDS AND PHRASES: store, storage, storage allocation, dynamic storage allocation, storage allocation strategy

CR CATEGORIES: 4.49

1. *Introduction*

Dynamic storage allocation systems can be classified into two types: those that move blocks around the store and those that do not. Systems of the second type may have many advantages but they are, except in special cases, inefficient in their use of store. In this paper we are concerned with making an estimate of this inefficiency. For simplicity we ignore most of the constraints which may affect the issue in a practical case and assume that:

(i) The total number of busy words may never exceed M.

(ii) The size of a block may never exceed n.

(iii) Any sequence of allocations and de-allocations which does not contravene (i) or (ii) is possible.

Graham [1] has discussed this problem in terms of a game between an attacker A and a defender D. A's moves have two phases: in the first he removes strings of squares (blocks of words) from the board (store) and in the second he forms new strings which D must place on the board in his move. He defines $N^*(M, n)$ as the smallest board on which D can force a win, and gives upper and lower bounds for $N^*(M, n)$. In this paper we strengthen these bounds and give the solution of the case where $n = 2$.

2. *Summary of Results*

THEOREM 1. $N^*(M, 2) = [(3M - 1)/2]$ *for* $M > 0$.

(We will consistently use square brackets to denote integer part.)

This theorem was a result of the hope that study of simple cases would provide a hint of a general solution.

THEOREM 2. $N^*(M, 2^a) \leq M(a + 1)$ *if M is a multiple of* 2^a.

* Programming Research Group, Oxford University Computing Laboratory

THEOREM 3. $N^*(M, 2^b) \geq 4\,Mb/13 - 16 \cdot 2^b/13 + 9\,M/13$.

These two results imply similar inequalities even when n is not a power of 2 since $n_1 > n_2$ implies $N^*(M, n_1) \geq N^*(M, n_2)$. Thus if $2^x < n < 2^{x+1}$, then

$$N^*(M, n) \leq N^*(M, 2^{x+1}) \leq M(x + 2) \qquad \text{if } M = k \cdot 2^{x+1},$$

and

$$N^*(M, n) \geq N^*(M, 2^x) \geq 4\,M(x + 2)/13 \qquad \text{if } M > 16 \cdot 2^x.$$

THE LIMIT $N'(n)$. The following inequality always holds for $M1$, $M2$, $n > 0$:

$$N^*(M1, n) + N^*(M2, n) + N^*(2n - 2, n) \geq N^*(M1 + M2, n),$$

since the following strategy cannot be defeated:

Divide the store into three independent regions with sizes $N^*(M1, n)$, $N^*(M2, n)$, $N^*(2n - 2, n)$ and follow in each of these the relevant optimum strategy. The overall strategy directs each block into some region without violating the restrictions applicable to that region. [To see that this is always possible suppose that a block of size x cannot be directed into any region. The regions must contain $M1 - x + 1$, $M2 - x + 1$, and $2n - x - 1$ words respectively. Hence collectively they contain at least $M1 + M2 + 2n + 1 - 3x$ words, and the new block would bring this up to $M1 + M2 + 2n + 1 - 2x \geq M1 + M2 + 1$ (since $x \leq n$).]

If we define $N''(M, n) = N^*(M, n) + N^*(2n - 2, n)$, the inequality becomes

$$N''(M1, n) + N''(M2, n) \geq N''(M1 + M2, n),$$

from which we can conclude that $N''(M, n)/M$ tends to a limit $N'(n)$ as M increases, but

$$\begin{aligned} N^*(M, n)/M &= N''(M, n)/M - N^*(2n - 2, n)/M \\ &= N''(M, n)/M + o(M). \end{aligned}$$

Therefore $N^*(M, n)/M$ also tends to $N'(n)$ as M increases. The results stated above give

$$N'(2) = 3/2,$$

$$4a/13 + 9/13 \leq N'(2^a) \leq a + 1,$$

$$4([\log_2(n)] + 2)/13 \leq N'(n) \leq [\log_2(n)] + 2.$$

These inequalities suggest that $N'(n)/\log_2(n)$ tends to some limit as n increases, or at least as n increases through powers of 2. There remains of course the possibility that it oscillates between 4/13 and 1.

3. *The Case* $n = 2$

THEOREM 1. $N^*(M, 2) = [(3M - 1)/2]$ *for* $M > 0$.

The proof consists of two parts. The first shows a defensive strategy which will win on a board of size $[(3M - 1)/2]$. The second shows an attacking strategy which will win on a board of size $[(3M - 1)/2] - 1$.

(1) D's strategy is: Never place a block of size 1 on any of the squares 2, 5, 8, $\cdots$, $3z + 2$ where z is the greatest integer such that $3z + 2 < [(3M - 1)/2]$.

The number of such squares is $[[(3M - 1)/2]/3] = [(M - 1)/2]$. The number of squares on which D may place a block of size 1 is $[(3M - 1)/2] - [(M - 1)/2] = M$. D can always place a block of size 1, given to him by A, on board. Suppose D is unable to place a block of size 2. D cannot place it on the pair $(3i - 2, 3i - 1)$ or on $(3i - 1, 3i)$, $(i = 1, \cdots, [(M - 1)/2])$. Therefore each triple $(3i - 2, 3i - 1, 3i)$ $(i = 1, \cdots, [(M - 1)/2])$ contains two busy words at least. Therefore $(1, \cdots, 3[(M - 1)/2])$ contains $2[(M - 1)/2]$ busy words. If M is odd this region contains $M - 1$ busy words. If M is even $M - 2$ busy words are contained in $(1, \cdots, (3M/2) - 3)$ and another must be contained in $((3M/2) - 2, (3M/2) - 1)$. For any M there are $M - 1$ busy words. A cannot form a block of size 2.

This completes the first part of the proof.

(2) A's strategy is:

(i) Form M blocks of size 1.

(ii) Select alternate blocks from these M (starting from the first) and remove such of these as are not adjacent to a gap whose size is odd.

(iii) Form as many blocks of size 2 as possible with the removed words.

The number of blocks selected in the first part of (ii) is $[(M + 1)/2]$. Suppose the number of odd gaps is x. The number of words removed is greater than or equal to $[(M + 1)/2] - x = [(M - 1)/2] - x + 1$. The total size of the gaps after (i) is $[(3M - 1)/2] - 1 - M$. Hence, $x \leq [(3M - 1)/2] - 1 - M = [(M - 1)/2] - 1$, and $[(M - 1)/2] - x > 0$. The total number of words comprised in blocks of size 2 which could be placed in these gaps is $[(3M - 1)/2] - 1 - M - x = [(M - 1)/2] - x - 1$. Step (ii) does not increase this quantity. D is unable to deal with the, at least, $[(M - 1)/2] - x$ words used in step (iii).

4. *The Upper Bound*

THEOREM 2. *$N^*(M, 2^a) \leq M(a + 1)$ if M is a multiple of 2^a.*

A defensive strategy using less than $M(2a + 2)$ words is easily shown: Blocks of sizes $2^i + 1, \cdots, 2^{i+1}$ are dealt with separately in a region of size $2M$ divided into areas of size 2^{i+1} each of which will take one block. If a block cannot go into this region, the number of occupied words in the region is at least that given by one block of minimum size in each area, i.e. $2M \cdot (2^i + 1)/2^{i+1} > 2M \cdot 2^i/2^{i+1} = M$. This shows that the strategy cannot be defeated. The absolutely rigid segregation of blocks into different regions according to size may lead to easy proofs but it is not certain to give ideal strategies. To what extent can we improve the strategy by slightly relaxing the rule? The defensive strategy given below achieves a 50 percent reduction in the area used by allowing blocks to go in regions whose natural purpose would be to contain smaller blocks.

PROOF. The defensive strategy is: To place a block of size S where $2^x < S \leq 2^{x+1}$, find the smallest integer k for which the squares $k \cdot 2^{x+1} + i$ $(0 < i \leq 2^{x+1})$ are free and place the block with its leftmost word on square $k \cdot 2^{x+1} + 1$. [In general the squares $k \cdot 2^x + i$ $(0 < i \leq 2^x)$ will be called a 2^x-area.]

The proof shows inductively that all blocks with sizes up to and including 2^n go into the first $\sum_{i=0}^{n} A_i$ words of the store where $A_0 = M$, $A_i = 2^i M/(2^i + 1)$, rounded up if necessary to a multiple of 2^a $(i \geq 1)$. The induction is based on the case $n = 0$ which is trivial. The theorem is proved by the case $n = a$ since $A_i \leq M$.

(This is where we use the fact that M is a multiple of 2^a.) The inductive step assumes that the result is proved for all $n' < n$ and that a 2^n block cannot be placed in the first $\sum_{i=0}^{n} A_i$ words of the store and shows that this can only happen when all M words are already in use. The area from $\sum_{i=0}^{m-1} A_i + 1$ to $\sum_{i=0}^{m} A_i$ is called the 2^m-block area and consists of an integral number of 2^x-areas for any $x \le a$. Suppose $m \le n$. Then the 2^n block cannot be placed in the 2^m-block area.

The 1-block area must contain at least 1 busy word in each 2^n-area. Each 2^n-area of the 2^m-block area contains part at least of a busy block of size S. By the induction hypothesis applied to $m - 1$, $S \ge 2^{m-1} + 1$ $(0 < m \le n)$.

If $S \le 2^n$, then the block is contained in a 2^p-area for some $p \le n$, which must lie within the 2^n-area. The density of busy words in such a 2^n-area is at least $(2^{m-1} + 1)/2^n$ $(0 < m \le n)$.

If $S > 2^n$, then for some p $(n < p \le a)$ the leftmost word of the block lies on a square $k \cdot 2^p + 1 = k' \cdot 2^n + 1$. Any such block, straddling x 2^n-areas, must lie within the 2^m-block area, and have a size at least $(x - 1)2^n + 1$. The density of busy words in the x straddled 2^n-areas is at least $\{(x - 1)2^n + 1\}/x \cdot 2^n \ge (2^n + 1)/2 \cdot 2^n$, since $x \ge 2$. Hence, the density of busy words in the 2^m-block area is at least

$$\min\left(\frac{2^{m-1} + 1}{2^n}, \frac{2^n + 1}{2 \cdot 2^n}\right) \qquad (0 < m \le n),$$

but

$$\frac{2^{m-1} + 1}{2^n} > \frac{2^{m-1} + \frac{1}{2}}{2^n} = \frac{2^m + 1}{2 \cdot 2^n} \quad \text{and} \quad \frac{2^n + 1}{2 \cdot 2^n} \ge \frac{2^m + 1}{2 \cdot 2^n},$$

so that the density of busy words is at least $(2^m + 1)/2 \cdot 2^n$ $(0 < m \le n)$. The number of busy words in the 2^m-block area is at least

$$M \cdot 2^{-n}, \qquad m = 0,$$

$$\frac{2^m M}{2^m + 1} \cdot \frac{2^m + 1}{2 \cdot 2^n} = M \cdot 2^{m-n-1} \qquad (n \ge m > 0).$$

The total number of busy words in the 2^m-block areas $m = 0, 1, \cdots, n$ is at least

$$M(2^{-n} + 2^{-n} + 2^{-n+1} + \cdots + 2^{-1}) = M.$$

This completes the proof of Theorem 2.

5. *The Lower Bound*

THEOREM 3. $N^*(M, 2^b) \ge 4\,Mb/13 - 16 \cdot 2^b/13 + 9M/13$.

A simple strategy for A would be to use $b + 1$ moves on the ith of which he first removes as many words as possible without creating a gap of size 2^{i-1} or greater, and then forms as many blocks of size 2^{i-1} as possible. This ensures that D can never use the same square twice for different blocks. The actual strategy described below is similar but more restricted in its removals so that it has the above property and some others which it does not share with this simple strategy.

A square is "captured" by A when either a block is placed on it by D or it becomes clear that D will never be able to place a block on it.

If photographs were taken of the board after each of D's moves, the $b + 1$ photographs would include almost $M(b + 1)$ pictures of squares covered by blocks. These are known as "word pictures."

When a square is captured, the responsibility for capturing it is shared between certain word pictures in such a way that nearly every word picture receives credit for at least 4/13 of a square. This gives a lower bound for the number of squares captured and thus for the number of squares on the board.

PROOF

(1) The attacking strategy which will win against any smaller store has $b + 1$ moves each of which uses blocks of only one size, namely 1 on the first move, 2 on the second, 4 on the third, and in general 2^{i-1} on the ith. A's strategy on the second phase of his moves is simple: Use as many blocks of the correct size as the number of available words will permit. The number of words unused after the ith move is thus less than 2^{i-1} and the total of such words is less than

A: $$\sum_{i=1}^{b+1} 2^{i-1} < 2^{b+1}.$$

A's strategy in the first phase of his moves depends on the following definitions. Suppose G is the gap between two adjacent blocks and that the distance between their centers is d, and that A's next move will use blocks of size 2^i. G is an *interior* gap if $d < 2^{i-1}$. A gap is *exterior* if it is not interior. A block is (L- or) *R-removable* if the gap on its (left or) right is interior. A block is *removable* if it is L-removable and R-removable. A word is *removable* if it is part of a removable block.

A's strategy in the first phase can be any which obeys the following rules:

Rule I. Remove only removable blocks.

Rule II. Do not remove two adjacent blocks in the same move.

Rule III. In each move remove at least half the removable words.

There is at least one strategy which obeys these rules since one of the two sets of alternate removable blocks must contain half the removable words.

If we consider only the states of the board at the beginning of A's moves we see that rules I and II mean that the only changes in state possible for a square of the board are those shown in Figure 1. To prove this, since clearly no other changes can take place except through the removal of a block, we look at such a removal. Suppose $B2$ in Figure 2 is removed in the move which uses blocks of size 2^i. Then $G1$ and $G2$ are interior gaps, $d1 < 2^{i-1} < d2$, $d1 + d2 < 2^i$.

So since neither $B1$ nor $B3$ can have been removed, D cannot put a block in the gap left by $B2$ and this gap will be interior at the start of A's next move.

This shows that no changes in the state of a word of store can take place against the arrows of Figure 1 and also that a word cannot be used for two different blocks at different times.

(2) If a word W is in use after D's ith move then (W, i) is called a *word picture* or the picture of W at that time.

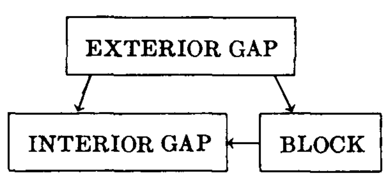

FIG. 1

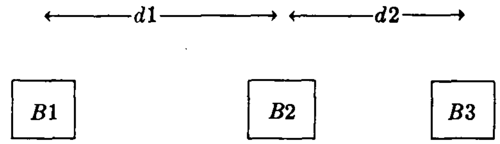

FIG. 2

From the note at A the total number of word pictures is greater than $(b + 1)M - 2^{b+1}$. A relationship between these word pictures and areas of the board provides the proof of the theorem. (According to the terminology of the informal sketch above, responsibility for the capture of the squares of the area is divided equally between the word pictures related to the area.) The areas involved are (i) areas on which blocks are placed, (ii) areas which at the beginning of some move are external gaps, upon which no blocks are placed, and which become internal at the end of the move; or areas which are external gaps at the end of the game. Figure 1 and the the argument concerning it show that an area can only be involved once in any of these ways.

(2.1) The area used by a block B of size 2^k is related to:

(i) The pictures of the words of B at the end of the move using 2^k blocks.

(ii) The pictures at the same time of up to 2^k removable words not removed in the first phase of A's last move. (These pictures are to be chosen so that none are related to more than one area used by such a block B, but otherwise chosen arbitrarily. The full 2^k are to be chosen if this many pictures remain unremoved that were removable, otherwise all those that do remain.)

(iii) Two of

(a) the pictures of the words of B at the end of the move using 2^{k+1} blocks (unless B is removable at the start of this move);

(b) the pictures of the words of a smaller block, adjacent to B on the left, from the third move after its creation (i.e. the move which used blocks 4 times its size) to the present move (using 2^k blocks) inclusive (provided that the smaller block became removable at the end of the present move);

(c) same as (b) but a block on the right.

(It is impossible for all three (a), (b), and (c) to be applicable since if (b) and (c) apply B is surrounded by removable blocks and so by interior gaps and so is itself removable.) (i), (ii), and (iii-a) all involve at most 2^k word pictures each. If (iii-b) [or (iii-c)] involves X word pictures and the other block has size 2^m,

$$\begin{aligned} X &= 2^m(k - m - 1) \\ &= 2^k \cdot 2^{-(k-m)}(k - m - 1) \\ &= 2^k \cdot 2^{-Y}(Y - 1), \qquad \text{where } Y \text{ is an integer} \geq 1, \\ &\leq 2^k \cdot \tfrac{1}{4}. \end{aligned}$$

Thus the area used by B is related to at most $13/4 \times 2^k$ word pictures.

(2.2) If a gap G of size g is included in the description in (2) and if the last move at the start of which it is exterior is that using 2^a blocks then G is related to the pictures of the words of the blocks adjacent to it from the third move in which they existed until the move using 2^a blocks inclusive.

If G is surrounded by blocks of sizes 2^b, 2^c then the number of word pictu related to G is $2^b(a - b - 1) + 2^c(a - c - 1)$.

Since G is not interior at the beginning of the move using 2^a blocks,

$$\begin{aligned} g &\geq 2^{a-1} - \tfrac{1}{2}\, 2^b - \tfrac{1}{2}\, 2^c, \qquad \text{by the definition of an interior gap,} \\ &= \tfrac{1}{2}\,(2^{a-1} - 2^b) + \tfrac{1}{2}(2^{a-1} - 2^c) \\ &= \tfrac{1}{2}\, 2^b(2^{a-b-1} - 1) + \tfrac{1}{2}\, 2^c(2^{a-c-1} - 1) \\ &\geq \tfrac{1}{2}\, 2^b(a - b - 1) + \tfrac{1}{2}\, 2^c(a - c - 1), \qquad \text{since } a - b,\ a - c \text{ are positive integ} \\ &= \tfrac{1}{2}[2^b(a - b - 1) + 2^c(a - c - 1)], \end{aligned}$$

i.e. the number of word pictures related to G is at most $2g$.

B: Therefore the number of word pictures related to any area by (2.1) or (2 does not exceed 13/4 times the size of the area.

(3) The number of word pictures not related to any area is less than 2^{b+1}

PROOF. Suppose (W, n) is a word picture and word W is part of a block Then either (a) B has been created in the nth move, or (b) B existed at the st of the nth move and was not removable, or (c) B existed at the start of the move and was removable.

If (a) then (W, n) is related to the area covered by B by (2.1-i).

If (b) then either B was formed on the $(n - 1)$-th move, in which case $(W,$ is related to the area covered by B by (2.1-iii-a), or B was formed earlier.

Now either (i) B will eventually become removable, or (ii) B will not eventu become removable. (i) can happen because a larger block is placed adjacent t or because a gap adjacent to B becomes interior. (W, n) is related to the are the block or the gap by (2.1-iii-b) or (2.1-iii-c) or by (2.2). If (ii) then B is jacent to a gap which remains exterior and (W, n) is related to the area of this by (2.2).

If (c): If there are R words in blocks of this kind at the end of the nth m then at least R words were removed at the beginning of this move (rule I Therefore at least $R - 2^{n-1} + 1$ of the R words were related to some area (2.1-ii): For fixed n the number of word pictures (W, n) not related to any are less than 2^{n-1}; altogether the number of word pictures not related to any are less than $1 + \cdots + 2^b < 2^{b+1}$.

The number of word pictures which are related to some area is greater t $M(b + 1) - 2^{b+1} \cdot 2$. Hence the inequality stated at B becomes $M(b + 1)$ $4 \cdot 2^b \leq 13A/4$, where A is the total area associated with word pictures; $M($ $1) - 4 \cdot 2^b \leq 13B/4$, where B is the size of the board. This can be strengthened the observation that the M squares used by D for blocks of size 1 on the move are related to only $2M$ word pictures at most since only clauses (i) (iii-a) of (2.1) can apply. Applying the inequality B to the remaining area, we h

$$M(b + 1) - 4 \cdot 2^b - 2M \leq 13\,(A - M)/4 \leq 13\,(B - M)/4,$$

$$B \geq 4\,Mb/13 - 16 \cdot 2^b/13 + 9M/13,$$

$$N^*(M, 2^b) \geq 4\,Mb/13 - 16 \cdot 2^b/13 + 9M/13.$$

ACKNOWLEDGMENTS. The author is indebted to M. D. McIlroy and to the referees for many helpful comments.

REFERENCES

1. GRAHAM, R. L. Bounds for dynamic storage allocation strategies (unpublished).

RECEIVED NOVEMBER, 1969; REVISED SEPTEMBER, 1970